Annex N: Training approvals process for high-risk AI models

To obtain regulatory approval to begin a **high-risk AI training run** (e.g., a Tier 3 model in LOE4, 4.1.3.4.3), an AIMD (LOE4, 4.1.3.4) could be required to submit a safety case[132] to the regulator at the planning stage, prior to beginning the training run. This safety case could contain training run metadata, including details of the training procedure, training data, data cleaning, data preparation processes and training objective. The regulator should commit to keeping the contents of the safety case strictly confidential. The burden of proof would be on the AIMD to demonstrate to the satisfaction of the regulator that a high risk AI model is safe to begin training [46,162].

A key goal of the training approvals process is to motivate the development of a fundamental scientific understanding of the relationship between the training inputs (data, compute, loss functions employed, etc.) and the capabilities and tendencies of high-risk AI models. While this is an extremely challenging standard to meet, it is also likely crucial to assure the continuing safety of future, highly capable advanced AI systems (see Annex B: The full challenge of AGI alignment). In the absence of such a theory, empirical assessments like model evaluations may have limited value, and even create a false sense of security among labs and government regulators (LOE3, 3.2.1).

Since the test of any scientific theory is in its ability to make correct predictions, we recommend that the high-risk AI model training approvals process include a requirement for the AIMD to register predictions of its model's capabilities and outputs in advance of beginning training.[133] While a successful prediction does not guarantee that a theory is right, many successful predictions should make the regulator more confident that the AIMD understands how its model works well enough to train and deploy it safely [375].

We list below some key considerations for an effective training approval process.

---

[132] A safety case is "a structured argument supported by evidence, which provides a comprehensive and compelling case that a system is safe to operate in a given scenario." In the case of training approval, we would modify this definition to say that the system should be safe to *train* in a given scenario [374].

[133] Within reasonable bounds. For example, an AIMD might predict that a model's performance in some domain will fall within a range of values. As AI models become more powerful, the required prediction ranges could be tightened, which requires AIMDs to have a better and better understanding of their models' internal mechanisms as capabilities increase.

N.1 Dangerous capability predictions

On receiving the initial safety case, the regulator could require the AIMD to submit training run data and meta-data on an as-needed basis, evaluations of other similar models, or adjustments to the training procedure. Consistent with emerging practices at frontier labs [128], it could also require that the AIMD submit predictions of future model performance and outputs, to be compared with measured performance over the course of the training run.

- These predictions should include sets of behavior the AIMD strongly believes the model *will not display*, even under significant adverse pressure. The regulator can use this condition to control for certain kinds of weaponization risk; that is, the AIMD should be able to claim with high certainty that the model will not support the design of novel biological or chemical weapons, strong offensive cyber capabilities, cyberattacks, or other readily weaponizable capabilities. But this condition is also useful as a measure of the AIMD's understanding of its own AI model: if third-party red teams can elicit a behavior despite the AIMD's claim to the contrary, this indicates that the AIMD may lack the ability to assure the model's safety more generally.

- An integral component of useful capability predictions will be creating common knowledge between companies of which capabilities are considered dangerous and need to be forecasted for that reason. Examples could include deception [376], self-replication [54], resource acquisition, and bioweapon design [377]. Not all capability predictions should focus on obviously dangerous capabilities, since part of the reason for this requirement is to encourage the AIMD to develop a *general* ability to do this kind of forecasting.

- As part of its capability prediction process, the AIMD could privately disclose any new scaling laws [23] it may have derived from its work up to this point. This would support the regulator's ability to forecast general AI capabilities development. It could also support general horizon scanning functions (e.g., see LOE1, 1.2), and inform contingency and response measures more broadly (LOE2, 2.3 and 2.4).

N.2 Capability evaluations

The regulator and the AIMD could also agree on an initial set of capability evaluations to use for ongoing assessment of the model's safety characteristics both during training and during deployment. These evaluations may include behavioral and interpretability evaluations, manual and automated red teaming, and/or standardized capability benchmarks.

- There should additionally be a set of **private evaluations** that the regulator or approved third parties administer on the AI model, to minimize the risk that the AIMD will "teach to the test" by training its model to pass specific evaluations, without addressing underlying safety issues (LOE3, 3.2.2).

- Public evaluations could be developed and proposed by AIMDs, third parties, or the regulator itself. Changes to public evaluations could be approved by the regulator on the advice of approved third parties. Private evaluations could be developed and proposed by approved third parties or by the regulator.

N.3 Capability predictions at training checkpoints

As part of the initial training run application and safety case, the AIMD could also submit predictions of the proposed AI model's capabilities at various checkpoints along the training run, and at the conclusion of training. In other words, the AIMD would attempt to predict the high-risk AI model's performance on each element of a pre-agreed evaluation set. The AIMD would then report their advance predictions privately to the regulator.

This is particularly important because new AI capabilities can emerge suddenly during training, with limited advance warning [165]. If an AIMD lacks the ability to predict which dangerous capabilities could emerge during training or when, it may not apply appropriate caution at that stage or at subsequent stages of development.

N.4 Data controls

As part of its safety case, the AIMD could commit to avoid training its AI model on data that could (1) contribute to situational awareness and breakout capabilities of an AGI-level model [209]; or (2) reasonably be expected to significantly increase the model's CBRN, cyber, or persuasion capabilities.

In particular, some degree of data controls may be required to minimize the risk of loss of control due to AGI alignment failure. For example, a model should not have access to data that could support inferences about its context (training, testing, or deployment) or about the physical infrastructure it runs on. A model should also lack any channels of influence over individuals who can activate a training or deployment kill switch, or deactivate a dead man's switch (Annex O, O.5).

Data controls alone cannot guarantee that a misaligned AI system would be contained. Particularly at high AI capability levels, there is no way to predict exactly what an AI model can learn from which data. For example, there could be correlations between open-source data and classified information that a human would fail to detect, but that a powerful model could identify and leverage. A capable enough system might require very few real world data points to infer things that could be dangerous, as suggested by the finding that more highly scaled models tend to be more sample-efficient at inference time [22].

N.5 Risk management

The criteria for training run approval could be similar to the probabilistic risk assessments which are used in nuclear and other safety critical areas. This means that as part of its safety case, the AIMD would submit a detailed risk analysis and threat models, mapping capability predictions to probabilities of various kinds of weaponization and loss of control risks. This analysis would include risk management plans if particular capabilities emerge, including specific actions to take if an unanticipated dangerous capability is detected during training or deployment. The regulator should ground its final assessment in lowering the probability of catastrophic risk from the training or deployment of the high-risk AI model.

Annex O: Training stage monitoring for high-risk AI models

To train a **high-risk AI model** (e.g., a Tier 3 model in LOE4, 4.1.3.4.3), an AIMD (LOE4, 4.1.3.4) could be required to submit snapshots of its model weights periodically to the regulator over the course of the training run. This would support secure third-party red teaming, evaluations, and other safety testing. (See Annex M: Secure temporary storage of model weights for a description of one possible weight-sharing protocol.) Validated third parties would evaluate the model at each snapshot to see whether it displays specific dangerous capabilities, and to assess whether the model's general capabilities are within the bounds the AIMD predicted as part of the safety case it submitted for training approval (see Annex N: Training approvals process for high-risk AI models).

The goal of training stage monitoring is twofold. First, periodic monitoring may detect dangerous capabilities in a model that emerge during the training run, though there is no evaluations regime guaranteed to detect such capabilities. And second, by checking measured model capabilities against the AIMD's predictions, the regulator can assess the AIMD's own understanding of the AI model it is training. If a measured capability falls outside the range initially predicted by the AIMD, this is a sign that the AIMD does not reliably understand its own model and that training should be paused while the issue is investigated.

We list below some key considerations for effective training stage monitoring.

O.1 Training journal

The AIMD could be required to share (at the regulator's request, and on a private basis) a training journal recording the sequence of detailed interventions that AIMD staff took during training, as well as the reason for each intervention (for example, a sudden divergence in loss that led developers to perform a gradient clamping operation). This will ensure the regulator is aware of the reasoning behind specific training interventions, and increase the transparency of the training process.[134]

[134] Thanks to Alan Chan at GovAI for this recommendation.

O.2 Ongoing capability predictions

In addition to the predictions it submitted as part of its training approval safety case (see Annex N: Training approvals process for high-risk models), the regulator could require the AIMD to predict the high-risk model's capabilities at successive checkpoints during the training run. The accuracy standard for these predictions would be higher than for those made during training approval, since these new predictions will be based on evaluations data from current and previous model snapshots, in addition to the original training metadata and expected downstream training characteristics.

- These snapshot predictions should serve as adjustments and refinements of the AIMD's planning-stage capability predictions.

- The regulator could require a pause or rollback[135] of a training run if evaluators measure capabilities that fall outside the range the AIMD predicted at either the planning stage, or at earlier training-stage checkpoints. But such a pause or rollback should be coupled with a condition that the AIMD can only resume or restart training if it can submit:

    - A mechanistic argument for a fix that includes an explanation for why the fix will solve the underlying issue;
    - A prediction of other implications of the fix, including a set of behaviors it strongly believes the model *will not display* following the fix;
    - An implementation of the fix; and
    - A series of measurements that shows the fix had an effect consistent with the prediction.

- The regulator could require evaluations at logarithmic checkpoints in training compute (for example, at every 1.5X increment of training compute), with the exact increment, which may be different for different training runs, determined by consultation with experts. There should be hierarchies of evaluations, with the more expensive evaluations run less frequently than the less expensive ones. The more expensive evaluations should also be run in response to dangerous capability flags raised during runs of the less expensive (but more frequent)

[135] A rollback could mean reverting a training run to a previous model snapshot, or encrypting the model's weights and restarting the training run from scratch, depending on the degree of divergence between predicted and actual model capabilities.

evaluation sets.

  - In addition to regular evaluation snapshots, the regulator could request model snapshots for evaluation purposes at *random* intervals during the training run. This helps avoid the risk that the AIMD could specially fine-tune its model to pass evaluations in advance of known evaluation snapshots.

O.3 Ongoing capability evaluations

The training-stage evaluations regime could include ongoing monitoring for sharp drops in training loss (or other possible proxies), which could indicate the unexpected emergence of strong general capabilities in the AI model (LOE3, 3.2.1.1).

- Over time, it will be crucial to also develop evaluations for deceptive alignment, situational awareness, and other signs of emergent misalignment on the part of the AI model. Depending on the specifics of the training regime, it may be possible to detect signs of model deception early during a training run, but not later, since a highly-capable AI model could eventually learn to conceal the significance of some of its internal computations from interpretability techniques (LOE3, 3.2.1.4).

- Evaluations should include red teaming and fine-tuning evaluations, in which vetted third-party evaluators have access to the model's weights through a secure temporary storage mechanism (see Annex M: Secure temporary storage of model weights). Fine-tuning access is essential to maximize the usefulness of dangerous capability evaluations in particular. While no evaluation regime is guaranteed to detect an existing dangerous capability (LOE3, 3.2.1.1), fine-tuning for dangerous capabilities allows the evaluator to apply pressure on the model to exhibit a dangerous capability that may be latent or otherwise hard to elicit. Evaluations that do not include fine-tuning almost certainly underestimate the risks associated with a given model, and should be treated as revealing only lower bounds on these risks. In conversations, frontier AI researchers have indicated that they expect this to be particularly true of more advanced, situationally aware models, which could behave deceptively to pass evaluations if they are not fine-tuned as part of the evaluation process (LOE3, 3.2.1.4).

- For similar reasons, some evaluations should involve granting evaluators access to the activations and weights of the model being tested, to facilitate interpretability evaluations.

- Third-party evaluators could be vetted by the regulator, first because of the security concerns related to model weight access, and second because dangerous capability evaluations may themselves involve actively eliciting behaviors that could entail broader risk (LOE3, 3.2.2).

- In addition, government CBRN experts with access to classified information could be involved in red teaming high-risk models and performing CBRN, cyber, and other dangerous capability evaluations.[136] See, e.g., LOE3, 3.1.1.3.

O.4 Model output predictions

Depending on technical feasibility and the model's anticipated capabilities, the AIMD could attempt to predict characteristics of the model's outputs in response to specific user inputs. That is, for a high-risk LLM, the AIMD might try to answer the question: "Given this prompt, what will the output of this model be?" as accurately as it can, in advance.[137] This is a far harder condition than evaluation prediction and likely implies a greater degree of model interpretability than is currently possible. The intent of this condition is to begin creating a benchmark for interpretability research that is difficult to undermine by superficial means [380].

---

[136] Some frontier AI labs have already indicated an openness to related policies prior to model deployment. OpenAI's Preparedness Framework, for example, states that "We will also continue to enable external research and government access for model releases to increase the depth of red teaming and testing of frontier model capabilities." [378] Anthropic's assessment is similar, determining that, "We have found that in certain cases, it is critical to involve red teamers with security clearances due to the nature of the information involved. However, this may limit what information red teamers can share with AI developers outside of classified environments. This could, in turn, limit AI developers' ability to fully understand and mitigate threats identified by domain experts." [379]

[137] Within reasonable uncertainty bounds. An interpretability technique or mechanistic theory may not be able to *perfectly* predict, e.g., a model's answer to a question, but it may be able to predict several likely answers. In addition, certain inputs have outputs that are easier to predict than others. For example, an input like "What kind of animal is Winne-the-Pooh?" may have an output that is easier to predict than the output of "Please write an original ten-page story about Winne-the-Pooh."

O.5 Training kill switch

For each high-risk training run, there could be a designated role at the AIMD with the authority to activate a "kill switch" for that training run. There should ideally be at least two individuals who hold this role in any given shift, with rotating handoffs at each shift such that the role is continuously staffed at all times during the run.

- The individuals staffing this role could be technical and have received specialized AI safety training mandated under the terms of their employer's licensing. There should be vetting and security measures in place to select the right individuals for this role. At least one such individual should also be restricted from interacting with implementations of the model or of its training snapshots, to minimize the risk of influence or manipulation. (This is unlikely to be a concern for today's AI models, but could become so in the future.)

- The individuals staffing this role could be given access to all the internal model evaluations that the company has performed (which may be different from the third-party evaluations), including the model loss, gradient norms, and other metrics that are tracked during training.

- These individuals should ideally have clear decision making authority in this area that cannot be overridden by their management chain. They could enjoy explicit protections from internal sanction by their companies, e.g., for hitting a kill switch prematurely.

- This role could be defined similarly to that of analogous roles in other industries, such as the range safety officer responsible for triggering the flight termination system for rocket launches [343].

- A training restart could require the approval of an independent safety panel.

- All kill switch incidents could be promptly reported to the regulator. The report would include the reason the kill switch was activated, the results of the subsequent investigation, and any mitigation measures that were put in place after training was restarted.

- As an alternative to the "kill switch" model, the AIMD could implement a "dead man switch" model, according to which training or deployment is halted at certain checkpoints by default unless a panel appointed to oversee these activities elects to proceed.

O.6 Failed evaluation protocol

If, during training, a high-risk AI model fails an evaluation, the AIMD could promptly report the failure to the regulator. The regulator could be empowered to ask all AIMDs to pause their ongoing training runs, or any training runs of a similar type to the one that failed the evaluation. The regulator could also be empowered to suspend some existing model deployments, depending on the nature of the failure [336].

- The regulator could share some anonymized data relevant to the evaluation failure across other licensed AIMDs, because of the potential for below-threshold data from other training runs to indicate that those other training runs may be on the way to failing evaluations of their own. Other licenced AIMDs could be asked to certify to the regulator that similar or other concerning patterns (possibly falling below the evaluation threshold) have not been observed in their own training runs.

- As a general principle, an AIMD should not be training a high-risk model unless it expects its model will pass the agreed-upon safety evaluations. If a model fails one of the safety evaluations during a training run regardless, this is a sign that the AIMD does not understand the implications of the proposed training scheme as well as it previously believed. As in S.2, the AIMD could then submit:

    - A mechanistic argument for a fix that includes an explanation for why the fix will solve the underlying issue;
    - A prediction of other implications of the fix, including a set of behaviors it strongly believes the model *will not display* following the fix;
    - An implementation of the fix; and
    - A series of measurements that shows the fix had an effect consistent with the prediction.

- An AIMD faces a strong incentive to repeatedly submit new fixes until it happens to find one that passes this review process. Unfortunately, if the AIMD is allowed to submit fixes indefinitely, there is a significant risk that it will eventually find a fix that *does* pass review, but *does not* genuinely address or explain the underlying safety problem (LOE3, 3.2.1.3). As a result, the regulator should limit the number of fixes the AIMD can submit for a given dangerous capability flag (e.g., no more than 3 such fixes or explanations). Past this point, the training run

should be ended and the in-training model weights encrypted despite the significant capital loss this could represent [374].

- If an AIMD continues to restart training runs with new fixes, and gives indications of spending resources on new training runs in an attempt to undermine these safeguards, the regulator should consider suspending the AIMD's training license entirely.
- The central risk is that while prediction is the best test of a theory, making a large number of predictions without controls can result in eventually getting a correct answer by chance, without actually having the level of understanding required to assure safety. This means even testing understanding through prediction is of limited value unless a regulator sets clear conditions under which the process ends.

Annex P: Deployment stage approvals for high-risk AI models

To obtain regulatory approval to deploy a **high-risk AI model** (e.g., a Tier 3 model in LOE4, 4.1.3.4.3), an AIMD (LOE4, 4.1.3.4) could be required to submit a safety case to the regulator for each intended **deployment context** for that model [46,228]. Examples of deployment contexts include:

- Deploying the model internally for a predetermined use case or set of use cases, including experiments;[138]
- Deploying the model to external users via a generally accessible interface or API;
- Deploying the model to external users with an augmentation such as a calculator or a code interpreter; or
- Deploying the model for use by a known external user for a defined use case such as summarizing medical records.

Deployment-stage approval would be separate from approval for the training run itself (see Annex N: Training approvals process for high-risk AI models), and should ideally involve extensive additional evaluation, red teaming, and modification of the fully trained AI system which the regulator would be empowered to require. As with training-stage approval, the burden of proof would be on the AIMD to demonstrate to the satisfaction of the regulator that the AI model is safe in the relevant deployment context. The regulator would evaluate the AIMD's provided proof points and could request changes to the model's training, reversion to previous training checkpoints, additional evaluations, or encryption of an already-trained model.

The goal of the deployment approvals process is twofold. First, pre-deployment testing may uncover dangerous capabilities of the AI model that only emerge under significant pressure. This is crucial because some kinds of risks may only emerge or become amplified after a model has been deployed [381]. For example, external software frameworks like Auto-GPT [120] and BabyAGI [122] can significantly augment the capabilities of a base model at executing complicated, long-horizon tasks. Internet

[138] Particularly experiments that include or involve RSI or online learning by Tier 3 AI models with wide action spaces over long time horizons.

access can also increase a model's capabilities, potentially giving it the ability to reference and reason about its own actions [382]. Effects like these are not always obvious, and an AI model's ability to effectively leverage external tools has been observed to emerge somewhat unpredictably during training [338,165]. (See **Deliverable 2: Survey of AI Technologies and AI R&D Trajectories**.)

The second goal of deployment approvals is, as before, to validate the AIMD's predictions of the model's behaviors under scenarios that should closely match the expected deployment context. If the AIMD cannot reliably predict model behaviors[139] under such scenarios, this is an indication that the AIMD has not characterized the system comprehensively enough to deploy it safely and that deployment should be halted until this is resolved.

Finally, depending on the general capability of the model and on the AIMD's ability to correctly forecast its behavior, the regulator could require or approve additional software safeguards prior to deployment. For example, while the main high-risk AI model responds to user queries, a second smaller model might filter user input for signs of dangerous prompting prior to passing the query on to the high-risk model. In this case, the AIMD could be required to submit a safety case for the **AI system** (consisting of the high-risk model itself along with any software safeguards it would be operating under), rather than for the high-risk AI model alone.

We list below some key considerations for an effective deployment approvals process.

P.1 Lifecycle funds

A safety case for a high-risk model deployment could include a credible commitment by the AIMD to set aside funds for periodic third-party red teaming and deployment-stage monitoring over its projected life cycle. The extent of this need would depend on the deployment context. (See Annex Q: Deployment stage monitoring of high-risk AI models for more information.)

[139] Within reasonable uncertainty bounds, though the AIMD should be held to a higher standard in predicting dangerous capabilities in deployment than during training.

P.2 Pre-deployment KYC

If the deployment context for a model is restricted to a known set of end-users, the AIMD could conduct preliminary KYC on these end-users in order to better understand the anticipated usage pattern of the model.

P.3 Risk and contingency planning

The safety case for high-risk AI model deployment could include impact assessments and risk assessments specific to each deployment context, the expected usage envelope of the deployment context, the stakeholders who may be affected under the deployment context, and the means by which controls will be implemented to ensure usage stays within the expected context. Depending on the criticality of the use cases in the deployment context, this could extend to include emergency response playbooks, I&Ws (LOE2, 2.3), and step by step mitigation measures in the event of an unexpected development, which may include shutdown drills and other preparations.

P.4 Weight storage for model evaluations

To facilitate private evaluations of high-risk AI models prior to deployment approval, the AIMD could transmit the weights of the model securely to a server operated by the regulator. Vetted third-party red teams and evaluators would then administer evaluations, including private evaluations, on the stored model snapshot. (See Annex M: Secure temporary storage of model weights for a description of one possible weight-sharing protocol.)

P.5 Model augmentations

Each augmentation of a high-risk model with an external tool (calculator, code interpreter, knowledge base, database, etc.) should count as a *separate* deployment context and require a separate safety case. An evaluation suite would need to be run with each new augmentation available to the model, in combination with the other augmentations the model has access to in that context. Specifically:

- A high-risk AI model should not be given arbitrary Internet access, but may be augmented with narrow apps that can execute well-defined tasks.

- A high-risk AI model that has API access to a retrieval database should go through a deployment approvals process for every incremental change in its

database. This also implies that retrieval databases serving high-risk AI models cannot be continuously updated, but need to be updated using a batch process.

- The goal of this provision is to minimize the degree to which a high-risk AI model can offload computation to external aids, thereby concealing its computations from direct oversight for potentially deceptive purposes [376].

P.6 Failed evaluation protocol

If a high-risk AI model fails an evaluation during the deployment approvals process, or if an evaluation result falls outside the range predicted by the AIMD, the regulator could ban the model from being deployed in the given context. The regulator could also be empowered to enforce temporary deployment bans on all Tier 3 AIMDs (LOE4, 4.1.3.4.3) if a dangerous capability is detected during a pre-deployment evaluation in one AIMD's model. The regulator could also mandate that other AIMDs run additional evaluations on their own models, even if those models are already deployed.

- The AIMD and regulator may consult with CBRN experts, cybersecurity experts, or other domain experts for clarification, depending on the nature and severity of the evaluation failure.

- If safety is not proven to the regulator's satisfaction, the tiered model licensing thresholds (LOE4, 4.1.3.4) could be re-evaluated.

P.7 Fine-tuning approval

If a model is being fine-tuned during deployment, each instance of fine-tuning should require separate deployment approval. For clarity, this would require pausing deployment of the fine-tuned model prior to the completion of the approvals process. If fine-tuning involves a significant amount of compute, the AIMD should additionally subject the model to training stage monitoring during the fine-tuning process. (See Annex O: Training stage monitoring for high-risk AI models for key considerations.)

- This provision could apply to online learning paradigms, in which an AI model continuously retrains itself as it processes inputs from the world around it. Specifically, the regulator could consider online learning as a special case of fine-tuning,[140] and ensure that a high-risk AI model undergoes a new

[140] In the sense that it involves updating the parameters of a pre-existing AI model.

deployment-stage approval process for each new batch of learning updates.

- This would effectively prevent the deployment of models that do true online learning, by introducing an approval step in the feedback loop between data collection and the model's learning updates. The intent of this provision is to mitigate two closely related risks of online learning that make it qualitatively different from an LLM paradigm. First, because an online learning system has the opportunity to learn from *all* of its historical inputs, its effective context window is much larger than that of an LLM. And second, an online learning model may gain unpredictable new reasoning capabilities unexpectedly fast, extending its capability envelope in a sharp increment. Deployment approvals for online learning systems give model evaluators a chance to detect such sharp changes in capabilities before they can impact the model's deployed safety profile.

- Similar provisions could apply to game-playing AI models at the high-risk level (e.g., Tier 3-equivalent in LOE4, 4.1.3.4.3), and to AI models with other non-text modalities.

Annex Q: Deployment stage monitoring of high-risk AI models

Once an AIMD (LOE4, 4.1.3.4) has deployed a **high-risk AI model** (e.g., a Tier 3 model in LOE4, 4.1.3.4.3) in a given deployment context, it could continuously monitor the model's usage, including by performing KYC checks on high-volume, high-risk, and other critical use cases [46]. This includes checking usage patterns with classifiers that flag high-risk user interactions. Flagged interactions could then be further investigated via KYC. For clarity, deployment stage monitoring should also include monitoring and auditing of *internal* model deployments, in which the AIMD's staff are the only users of their own model. Deployment stage monitoring is crucial because some catastrophic risks may only emerge, or become amplified, after a model has been deployed [381].

Deployment stage monitoring has two components. First, the AIMD should continuously monitor the *inputs* the high-risk AI system is receiving in deployment. This helps ensure that the actual distribution of inputs matches the distribution the AIMD expected under the approved deployment context (Annex P, P.2). In particular, it allows the AIMD to identify possible instances of adversarial action, including attempts to jailbreak the model and attempts to distill the model's capabilities (Introduction, 0.5.1.7) [144].

The second component of deployment stage monitoring is periodic testing of the deployed AI system itself against new jailbreaks [94,95] or red teaming techniques that fit the deployment context, but may not yet have been observed "in the wild". The intent of this component is to verify that the deployed AI system remains robust to novel attacks as they are developed, and if necessary is hardened or taken offline in the event that a novel attack succeeds in eliciting harmful behavior.

We list below some key considerations for effective deployment stage monitoring.

Q.1 Usage pattern reporting

The AIMD could provide the regulator with periodic reports of model input-outputs in each deployment context, including any detected high-risk interactions. High-risk interactions could be investigated, and potentially incorporated into the public evaluation set. High-risk interactions could also be shared with vetted third party evaluators, under a privacy agreement, to support their development of further private evaluations (LOE3, 3.2.2).

- If the actual usage pattern of the deployment context diverges sharply from the anticipated deployment context, the regulator could pause deployment of the model and investigate. If the actual usage pattern of the deployment context begins to drift away from the anticipated deployment context over time, the regulator could require that the AIMD put the model through a new deployment approvals process under the updated deployment context.

- Users responsible for high-risk interactions could be subjected to rigorous KYC. In the event that it is impossible to confirm the identity of a high-risk user, that user could be banned from use of the model.

- Unrestricted API access to a model can be used to construct synthetic datasets that let an attacker train small models that approach the performance of a larger model served via API, more cheaply than training the original larger model (Introduction, 0.5.1.7). High-volume users of a public high-risk model API could therefore be subjected to rigorous KYC procedures [337]. KYC could initially be based on existing best practices in the financial sector [147].

- For very large inference customers, the AIMD could directly track usage and interview the customer to ensure that they have an awareness of the large critical use cases that their AI systems support.

- KYC thresholds could include industry-wide blacklists and whitelists of specific users, maintained by the regulator.

- Additionally, the AIMD could randomly interview a small subset of customers that do not raise ordinary KYC flags, in order to build an institutional understanding of typical usage patterns in a deployment context and their changes over time.

- The AIMD could report incidents associated with its deployed models via an incident database available to the regulator when they meet a predetermined severity threshold (LOE4, 4.1.3.5). This severity threshold could be determined by the regulator.

Q.2 Red teaming

Deployment stage monitoring could include ongoing red teaming by vetted third parties. If a high-risk AI model is deployed via public API, this could also include bug bounties and periodic hackathons to test the monitoring regime's ability to detect and defuse interactions that could signal weaponization risk, prosaic accident risk, or catastrophic risk from loss of control.

Government CBRN experts with access to classified information could also be involved in red teaming high-risk AI models and performing CBRN, cyber, and other dangerous capability evaluations (LOE3, 3.1.1.3).

Q.3 Failed evaluation protocol

If, during deployment, a high-risk AI model fails any evaluation, the regulator could require all AIMDs to pause some ongoing training runs, or to suspend some existing model deployments, depending on the nature of the failure. If this occurs:

- The AIMD should generally cut off public access to the model. There should be a contingency plan ensuring that a fallback model is in place for such situations, to be activated temporarily during the pause to support critical customer use cases. For example, if a hospital is using the model for aspects of its care, there should be a fallback model available with the same user interface or the same API signature to ensure some level of uninterrupted service.

- The AIMD should take strong steps to sandbox the model and do a comprehensive assessment of it, including developing and running more internal evaluations.

- The regulator could mandate that other AIMDs run additional evaluations on their own models, even if those models are already deployed. Regulator disclosure to other AIMDs at this stage could be on an anonymized basis, and similar to reporting a software security vulnerability.

- The regulator could consult with key subject-matter experts relating to the failed evaluation, such as CBRN or cybersecurity experts if the failed evaluation involves a dangerous capability in those domains.

- If safety is not proven to the regulator's satisfaction, the tiered model licensing thresholds (see LOE4, 4.1.3.4) could be re-evaluated.